

A transformational approach to quantum field theory

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Inspired by the scattering transformation constructed by Stéphane Mallat [1], we develop a transformational approach to quantum field theory. It is based on a canonical (or Heisenberg approach), in contrast to the more common Lagrangian (Schrodinger or Feynman path integral) approach. A functional transformation of functionals of the fields and field momentums on the cotangent bundle, T^*M^n , to an analytic function on $\mathbb{C}^n$, iH , is constructed via a logarithmic generating functional, where n is the number of fields. The functional transformation is a projection of the motion onto the Renormalization Basis (or scattering matrix, S-matrix, basis), with basis vectors $|\beta(p, x)\rangle = |\beta(z = p + ix)\rangle$ on a Hilbert space that are solutions to the Renormalization Group Equations (RGEs). The transformation has generated a function, iH which is a linear generator of finite group actions of the Weyl-Heisenberg group, $\mathbb{H} = H \otimes dH = H \otimes \text{Ad}(H)$. For this reason we call it the Heisenberg Scattering Transformation (HST). There is nothing special about the Weyl-Heisenberg group, so this transformation will be generalized to a Group Scattering Transformation (GST), so that any additional Lie group symmetries can be embedded in the dynamical flow. Streamlines in the H Lie group direction are given by $\text{Re}(H) = \text{constant}$ and streamlines in the $\text{Ad}(H)$ group direction are given by $\text{Im}(H) = \text{constant}$. The dynamics has been reduced from diffeomorphic flow on the cotangent bundle to geodesic motion on an n -dimensional complex hyperplane, $\mathbb{C}^n$, of a Hilbert space, $|\beta(z)\rangle$. The equivalence classes of the topology of the analytic function $iH(\beta)$ (β_{0k} for the case of simple poles and zeros, but algebraic structures in general) are the ground states of the system. A novel perspective to quantization is developed based on the observation of the system at times separated by intervals much greater than the natural periods of the system given by $2\pi/(dH_2/d\beta)$ will be ergodic and can only be treated statistically. Since the phase of the system is cyclic, probability distributions must be quantized to ensure periodicity in the phase. Finally the coupling of one quantized system to others will be addressed by forming the cotangent bundle of the linear subspace of $\mathbb{C}^n$ extended to include the n_c coupling constants (that is $T^*\mathbb{C}^{n+n_c}$). This cotangent bundle is then transformed by the GST to $\mathbb{C}^{2n+2n_c}$, then a second quantization is done. Thermal equilibrium is shown to be $\exp(-\Delta H/k_B T)$ above each ground state, β_{0k} . Furthermore the thermal force is normal (the covector) to the geodesics (leaves of the foliation defined by $\text{Re}(H(\beta)) = \text{constant}$), that is the covector dH (with geodesics, leaves of the foliation defined by $\text{Im}(H(\beta)) = \text{constant}$). This is the coupling of the quantum system to superordinate quantum systems. The covectors associated with the coupling constants are the coupling of subordinate quantum systems into this system. Gravity will then be identified as the cosmological system of all systems with the equivalence classes of the ground states being different universes. The details of a practical implementation, applications to ϕ^4 theory, 1D stochastic Burger's Equation, 2D MHD pulsed power liner implosions, and a closed form solution via canonical variable transformation of a ion and a guiding center electron will be presented.

I. INTRODUCTION

A long standing practical and technical complication of field theory has been the process of regularization and renormalization of the theory. Constructing regularizations that isolate the singularities yet respect symmetries of the action has been challenging to do, and solutions have been found on an *ad hoc*, case-by-case, basis. The ability to renormalize the theory without changing the form of the action has been challenging and also only done on an *ad hoc*, case-by-case, basis. A change of the independent coordinate from the position to the momentum basis has only been of limited utility. Ken Wilson spoke about a more direct approach in his 1971 seminal paper in Physical Review [2]. When trying to block average the physics he longed for a transformation that was “*a quantitative characterization of a complete orthonormal set of minimal wave packets. For quantitative purposes one would have to take into account tails of the*

wave packets which extend outside their assigned cells. It will be assumed here that one can divide phase space into cells of unit volume in any way one pleases and still be able to construct a corresponding set of minimal wave packets. There is no guarantee that this is actually possible, and no examples of such a set of wave packets will be given here.” He then changed course to develop the renormalization group equations.

Recent advancements in the mathematics of signal and image analysis [1, 3–6] have given clues as to a transformational approach to quantum field theory by what Mallat calls a scattering operator or transformation of the signal or image. It turns out that this name is very well chosen. It will be shown that the correct formulation of the transformation is a change to the S-matrix (or renormalization basis) of the canonical approach to field theory. The application of what we call the Heisenberg Scattering Transformation (HST) to any functional of the field and/or field momentum is the generating functional

of the iH linear propagator of finite group actions of the Weyl-Heisenberg group. In fact, the HST is a mapping of functionals on the cotangent bundle T^*M^n , where n is the number of fields, to an analytic function, iH on $\mathbb{C}^n$. This is done via a logarithmic generating functional that is a projection of the motion onto the Renormalization Basis (or scattering matrix, S-matrix basis), with basis vectors $|\beta(p, x)\rangle = |\beta(z = p + ix)\rangle$, on a Hilbert space that are solutions to the the Renormalization Group Equations (RGEs). This is an n -dimensional complex hyperplane $\mathbb{C}^n$ of the Hilbert space. This is what is referred to in the Machine and Deep Learning (MLDL) literature as a Reduced Order Model (ROM) with coordinates $(\beta_1, \dots, \beta_n)$. The dynamical motion will be geodesic motion on this hyperplane with streamlines of this H group action being contours of $\text{Re}(H(\beta)) = \text{constant}$, and streamlines of the $\text{Ad}(H)$ group action (that is dH direction) given by $\text{Im}(H(\beta)) = \text{constant}$. In summary, diffeomorphic flow on the cotangent bundle has been reduced to geodesic motion on a n -dimensional complex hyperplane $\mathbb{C}^n$ of the Hilbert space of solutions to the RGEs. There is nothing special about the Weyl-Heisenberg group. This method can be generalized to any Lie group G . We call this generalization the Group Scattering Transformation (GST). The GST can be applied to any other Lie group symmetries of the dynamical motion, further foliating the dynamical manifold via the group product.

The HST is an iterative logarithmic generator that calculates H_{m+1} given H_m . The complex m -th order complex curvature can be calculated as $dH_m/d\beta$. Since $H_m = \ln(\beta)$, the complex curvature is $1/\beta$ (that is an inverse radius of curvature). If beta is written as $\beta = Je^{i\theta}$, then the real part of the curvature is $1/J$, and the radius of curvature is J . The complex part of the curvature is 1. This is the familiar expression for the frequency in the solution to the Hamilton-Jacobi equation (a canonical transformation to time invariant coordinates (J, θ) generated by the action), $\omega = \partial(\text{Re}H)/\partial J$. The frequency is $\omega = 1/J$ if $\text{Re}H(J) = \ln J$. This complex curvature $S_m = dH_m/d\beta = d^m H(\beta)/d\beta^m$ can be viewed as the S-matrix. The order of the S-matrix is equivalent to the number of particles that are interacting in the Hamiltonian H . In this respect, it is a Mayer cluster expansion. For instance, if the Hamiltonian is written with only binary interaction terms, the S-matrix will have only terms up to second order. But for the ϕ^4 theory, it will be shown, in Sec. VIII, that the S-matrix will only have terms up to fourth order. In Sec. VII, it will be shown that the first order term can be interpreted as the inverse of the action $1/\langle J \rangle(p, x)$ averaged over the stochastic or quantum fluctuations; and that the second order term can be interpreted as the harmonic oscillation frequency $\omega_0(p, x)$ of the ground state, or inverse of the mass $1/m(p, x)$. The size of the stochastic or quantum fluctuations will be $\sqrt{m}$.

Although the motion in this hyperplane will not be constant, it will be bounded with a variance of order

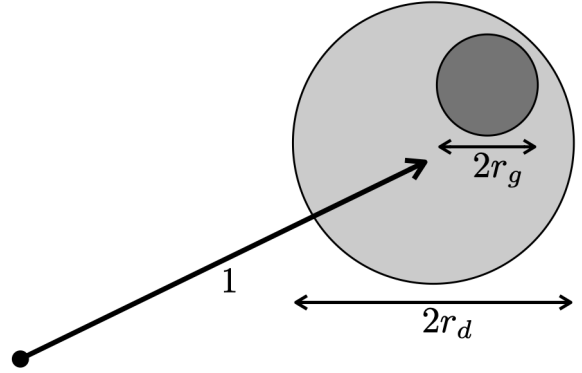


FIG. 1. Display of the precision of the cluster in $\mathbb{C}^N$ of the dynamical submanifold (r_d^2) and of the emergent behavior (r_g^2).

$(2\pi)^n$. When compared to the complete Hilbert space, this hyperplane will be of fractal dimension n/N , where N is the number of x -values on which the field is defined. This leads to the motion forming a cluster of precision n/N . Since the dynamical motion is determined by the analytic function $H(\beta)$, it is equivalently specified by the topology of the complex manifold. This topology is specified by the equivalence or homology classes of the topology, the ground states of the system (β_{0k} for the case of simple poles and zeros, but algebraic structures in general). These ground states are the emergent behaviors of the system if there are energy dissipating terms in the dynamics. If there are topological obstructions (that is topological invariants such as vorticity or helicity), the action will be prevented from going to zero for n_t linear combinations of actions in the ground states, where n_t are the number of topological invariants. Rather than β_{0k} being a point, it will be a loop. An ensemble of relaxed motions will have a variance $(2\pi)^{n_t}$, and will be a manifold of fractal dimension n_t/n with respect to the dynamical manifold. The cluster of the ground states will have a precision of n_t/N when compared to the Hilbert space and n_t/n when compared to the dynamic manifold.

This is shown in Fig. 1 for the case of $n = 6$, $n_t = 1$, and $N = 256^2$ (that is, 2D resistive MHD with vorticity as a topological invariant). The precision of the dynamical manifold will have a precision of $r_d^2 \approx 0.0001$, and the ground states will have a precision of $r_g^2 \approx 0.00002$ when compared the the Hilbert space and $r_g^2/r_d^2 \approx 0.2$ when compared to the dynamical manifold.

We emphasize that β are not constants of the motion, but the distributions of β can be used to discriminate and to identify dynamical systems, and to generate ensembles of the emergent behavior of dynamical systems. The raw values of the transform used in these applications have N components in (p, x) even though they span a small fraction of those dimensions as previously discussed.

In order to understand the topology of the dynamics

and therefore be able to predict the dynamics, further analysis of the dynamical topology must be done. For this analysis, one needs have an ensemble of field evolutions (technically functionals of the fields and field momentums, $F[\pi(x), f(x)]$) for different initial conditions and possibly different or changing coupling constants. This can be either experimentally measured or generated via computer simulation. The HST transformation then should be applied to the fields. A Principal Components Analysis (PCA) should then be done to identify and project the dynamics onto the complex n -dimensional hyperplane. The PCA vectors are the solutions to the RGEs. It should be noted that these solutions are complex valued, unlike the traditional solutions to the RGEs, which are real valued. Knowing the phase of the solutions is as important as knowing the modulus.

Given the ensemble of geodesic motion on this complex manifold, what needs to be estimated is the associated analytic function or the algebra of the topology. A good way to do this is with a Multi Layer Perceptron (MLP) with ReLU activation. It is well equipped for the task, since it is a piece-wise linear universal function approximator. The function that it is approximating is maximally flat, single valued, but does have singularities (that is discontinuities in the derivative). The MLP should have an encoder/decoder architecture as shown in Fig. 2. The number of nodes in the middle, encoded layer should be $2n + 1$, because the MLP will need to encode the PCA vector components of the HST into time invariant coordinates (J, θ) and the dynamical or true time τ . The time invariant coordinates are ones that satisfy the Hamilton-Jacobi equation

$$\frac{S(q, J, t)}{\partial t} + \text{Re } H \left(p = \frac{\partial S}{\partial q}, q, t \right) = 0 \quad (1)$$

which generates the transformation by using the action functional

$$S[q(t)] = \int p dq - H dt \quad (2)$$

as the generating function of the canonical transform, or equivalently finds coordinates such that $\partial(\text{Re}H)/\partial\theta = \partial(\text{Im}H)/\partial J = 0$. Practically, the canonical momentum nodes will be the curvature or frequency, $1/\bar{J} = \omega(J) = \partial(\text{Re}H)/\partial J$. The decoder will then advance the input state from 0 to τ and map the state back to $\beta(\tau)$. If the conventional or real Hamiltonian is not an explicit function of time, the decoder will simply advance θ by $\omega\tau$ and map the state back to $\beta(\tau)$. Note that if the conventional or real Hamiltonian is separable, (J, θ) are action-angle coordinates (with $J = \oint p dq$) and the motion is decoupled. The encoder/decoder layers will capture the graph of the algebra, that is the topology. This is by construction since the encoder/decoder are fully connected graphs with the connections optimized to capture the topology. The MLP is constructed to capture the solution of the Hamilton-Jacobi equation by the encoder/decoder architecture to/from the correct number of time invariant coordinates.

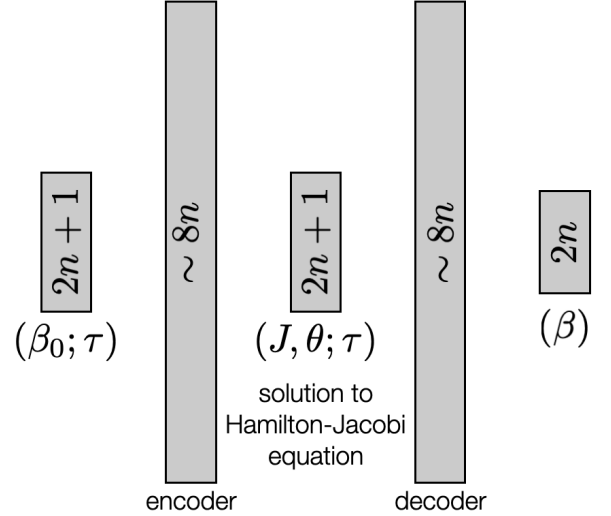


FIG. 2. Structure of the Multi Layer Perceptron (MLP) used to discover the topology of the dynamical manifold and predict the dynamics, equivalently to solve the Hamilton-Jacobi equation for the complex Hamiltonian. It has three hidden layers in a encoder/decoder configuration. It encodes the initial field coordinates β_0 and dynamical time τ to time invariant coordinates (J, θ) (that are the solution to the Hamilton-Jacobi equation) and dynamical time τ . It then advances the angle θ by $\omega\tau$, where $\omega(J) = \partial(\text{Re}H)/\partial J$, and decodes the coordinates into the field coordinates at τ , $\beta(\tau)$. The number of nodes in the encoder and decoder are approximately $8n$, but will vary depending on how close to singularities the phase space trajectories approach.

To build intuition, consider the two level sets of an analytic function, that are the contours of $\text{Re}(H(\beta)) = \text{constant}$ and $\text{Im}(H(\beta)) = \text{constant}$ as shown in Fig. 3 for the analytic function $H(\beta) = (\beta + 1/\beta)/2 = (\beta + i)(\beta - i)/2\beta$. They form two 1D foliations of the complex plane that are orthogonal and conformal. They define a diffeomorphism where the motion is along one set of leaves, with the motion clocked by the other set of leaves. Because of the orthogonal and conformal construction, the diffeomorphism conserves volume and is therefore unitary. Note the symmetry of the two sets of curves for H and dH . The motion can either follow the leaves of H and be clocked by dH or vice versa. This is the simple statement that $\text{Ad}(\text{Ad}(H)) = H$. This can be accomplished in the complex plane by rotating by $\pi/2$ or multiplying by i , or by interchanging π and f by the well known canonical transformation in phase space. Note that the motion along the level set of H (that is contours of $\text{Re}(H(\beta)) = \text{constant} = E$) is clocked by the parameter $\tau = \text{Im}(H(\beta)) = \text{constant}$. This is not the topology of $\mathbb{R}$, evidenced by the closed contours $\text{Re}(H(\beta)) = \text{constant} = E$ with topology of S^1 in Fig. 3, and the obvious topological obstruction or branch cut between $+1$ and -1 . (This is a two-sheeted Riemann surface, or manifold, with branch points $+1$ and -1 and a branch cut between those points. Both sheets cover the

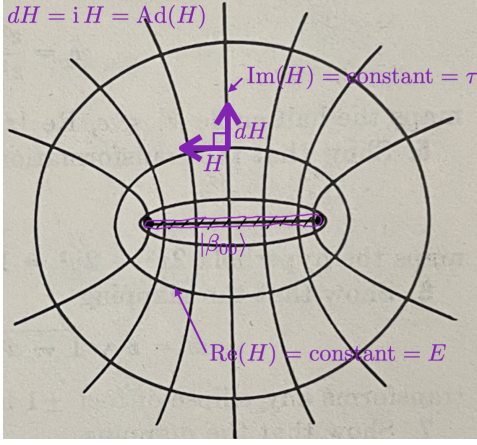


FIG. 3. Plots of the level sets of the function $H(\beta) = (\beta + 1/\beta)/2 = (\beta + i)(\beta - i)/2\beta$ showing the contours of $\text{Re}(H(\beta)) = \text{constant} = E$ and $\text{Im}(H(\beta)) = \text{constant} = \tau$. Equivalently, the group actions of H and dH . Shown also are the branch points at $+1$ and -1 , as black dots, and the branch cut between those points, as the hashed line. This is a two-sheeted Riemann surface (that is manifold) with sheets that cover the entire plane and are connected by the branch cut. The equivalence class or ground state, β_{00} , is the cycle around the branch cut. Specification of the branch cut, β_{00} , is equivalent to specifying $H(\beta)$, since $H(\beta)$ is the solution to Laplace's equation given β_{00} . This mapping is of importance in certain aerodynamical applications since it maps certain circles to Joukowski profiles which have the general character of airplane wing profiles.

entire plane. The equivalence class or ground state, β_{00} , is the cycle around the branch cut.) The trajectory can only be locally mapped to $\mathbb{R}$. The two sets of contours, the analytic function $H(\beta)$, define two complementary group actions and are inexorably intertwined into one group, $\mathbb{H} = H \otimes dH = H \otimes \text{Ad}(H)$, with one topology. Here dH is the exterior derivative of a zero form (function), H , on phase space with a symplectic metric $d\pi \wedge df$, or rotation by $\pi/2$ in the complex plane.

To quantize this theory, consider the case of observing the system at dynamical time, τ_0 , then predicting its state at a later time, $\tau_0 + \Delta\tau$. If $\Delta\tau$ is less than the natural periods of the system given by $2\pi/\omega_0$ (where $\omega_0 = dH_2/d\beta$, the second order S-matrix value), there is no problem in predicting the state of the system. This is the classical limit. Due to the non-linear character of the behavior, even if the system is integral, there will be exponential divergence of points on the submanifold on times greater than $2\pi/\omega_0$. This is the quantum limit where the system must be treated statistically. The system will remain close to the geodesic surface defined by $\text{Re}(H(\beta)) = \text{constant}$, but its position on the geodesic will be uniformly distributed in the distance along the geodesic (the phase angle of action-angle coordinates). Since this phase is cyclic, probability distributions must be quantized to ensure periodicity in the phase. So, the energy of any state can be written as $E = H_{0k} + \Delta H_{kn}$,

where $H_{0k} = \text{Re}(H(\beta_{0k}))$.

This is a subtle yet very significant departure from the traditional view of quantization. Each system is quantized at its own natural scale of action given by H_0/ω_0 not $\hbar$, and the uncertainty limit is in terms of this action scale. It has the ramification that there is now no difference between the stochastic uncertainty of nonlinear dynamics, quantum mechanical uncertainty, or the statistical uncertainty of kinetic theory (that is statistical and continuum mechanics). It also enables a systematic unified framework for all dynamical systems ranging from QCD, to QED, to atomic, to clock pendulums, to the weather, to gravity/cosmology, as will be discussed in the following paragraphs. The only difference is the time scale of the quantization that range from 10^{-18} to 10^{15} sec, as shown in Table 4. This figure also shows the lower uncertainty bound on the action.

The introduction of $\hbar$ into conventional quantum theory comes from a conflation of the observed time t with the dynamical time τ , which for the hyperbolic Minkowski metric is $\tau^2 = t^2 - (x^2 + y^2 + z^2)/c^2$. For systems with very large ground state frequencies so that the particle is moving at relativistic velocities near the speed of light (such as QED, QCD, black holes, and highly ionized large Z atomic systems) there is a practical problem in observing the quantity $\pi = \Delta f/\Delta\tau$ because of the finite speed of light. That is to say there is a lower bound to the action that can be observed from the rest frame (space-like interval) given by $\hbar$. (It should be noted that this limit does not exist if the action is observed from the frame of the particle. This fact is the origin of the paradoxes, such as the EPR paradox, associated with the conventional view of quantum mechanics.) Consequently, the system must always be observed at $\Delta\tau \gg 1/\omega_0$, so that it only can be observed in the quantum regime, where there is irreducible uncertainty in estimating canonically conjugate variables (that is the conventional Heisenberg Uncertainty Principle). Values of the characteristic actions that are less than $\hbar$ are indicated in bold in the figure. For these systems the conventional uncertainty principle applies, and they can never be classically observed from the stationary space-like frame, but they can still be classically (deterministically) observed from the frame of the moving particle. **Need to get this more solid with both $1/\omega_0 = m_{field}$ and $S_0 = H_0/\omega_0$ and the $m_{field}/m_0 = \gamma \gg 1$ relativistic limit where the system can not be viewed classically from the rest frame, only from the frame of the particle. Also need to talk about the topological obstruction inducing a rest mass m_0 .**

To couple the quantum system to superordinate and subordinate quantum systems, a second quantization must be done. This is done by forming the cotangent bundle of the linear subspace of $\mathbb{C}^N$ (that is $\mathbb{C}^n$) extended to include the n_c coupling constants, $T^*\mathbb{C}^{n+n_c}$. Apply the GST to transform $T^*\mathbb{C}^{n+n_c}$ to $\mathbb{C}^{2n+2n_c}$, then quantize. The superordinate system, if assumed to be large and in thermal equilibrium with a thermal bath

<u>Field theory</u>	<u>$1/\omega_0 (s) = \Delta\tau$</u>
gravity	10^{15}
weather	10^5
clock	10^0
atom	10^{-9}
QED	10^{-15}
QCD	10^{-18}

FIG. 4. Table of quantum scales $\Delta\tau = 1/\omega_0$ for several well known systems. For scales less than 10^{-9} , it becomes relativistically impossible to make differential measurements at those scales.

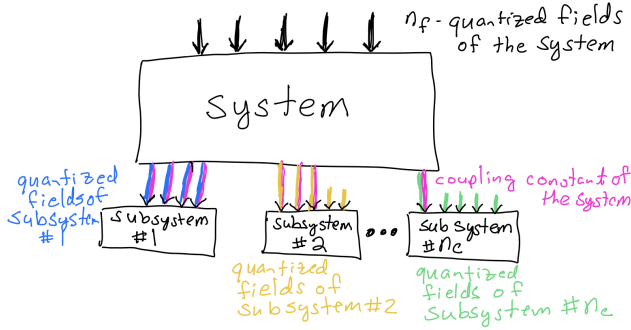


FIG. 5. Coupling of the quantum system to superordinate and subordinate quantum systems.

with temperature T , will evolve the statistical distribution in the quantized dH direction (the covectors associated with the fields), with a transition kernel $k(kn, k'n')$ that satisfies detailed balance and has a thermal equilibrium of $\exp(-\Delta H_{kn}/k_B T)$ above each ground state, β_{0k} , and $\exp(-H_{0k}/k_B T)$ for the relative population of each complex. The thermal force is normal to the geodesics (that is, normal to the leaves of the foliation) defined by $\text{Re}(H(\beta)) = \text{constant}$, dH . The quantum system is coupled to the subordinate systems via the quantized dC direction, whose quantum levels and topology are determined by the first quantization of the subordinate systems. A diagram of this arrangement is shown in Fig. 5. This is the basic building block of a directed graph of systems.

One can also view the superordinate coupling from the perspective of extending phase space into the $\text{Ad}(H)$ di-

rection as is done when a resistive force is added to a Hamiltonian system [7]. This is extending the dynamics to the product group $H \otimes \text{Ad}(H)$. The extended Poincaré one-form would be $\lambda = \pi df - Hd\tau$ and the Poincaré two-form or the symplectic metric would be $d\lambda = d\pi \wedge df - dH \wedge d\tau$, explicitly demonstrating the the $\text{Ad}(H)$ group is in the dH direction. It should be noted that the superordinate coupling is not in any arbitrary direction in the Hilbert space, it is restricted to dH , which is an n -dimensional linear submanifold of the Hilbert space. The first quantization of H yields the particle of the field. The second quantization of $\text{Ad}(H)$ yields what is traditionally called the gauge boson of the field.

The system, without the subordinate and superordinate couplings, is a stationary system that is translationally invariant. Since the x coordinate is often one dimensional and identified with time by extending the phase space, this means that there is time reversibility – there is no direction to time. This changes when there are either subordinate or superordinate couplings (or both). The time dependent changes to the coupling constants and fields, by definition, make the evolution of the system non-stationary. The resulting lack of time translational invariance gives a direction to time, that is makes the dynamics irreversible.

Gravity can now be brought in as the cosmological quantum system of all quantum systems. All fields from all systems are the coupling constants of this cosmological system. Assume that there is a single cosmological field with ground states, β_{ck} , that are simple poles and zeros (x-points and o-points), and a Hamiltonian of simplicity or organization (long range attraction of systems). The equivalence classes of the ground states can be identified as universes. The poles or x-points which we will call the Asgard, are points at which small changes can move the cosmological quantum system from one universe to another. The Asgard are the Stargates between universes. If the system is in the universe of an Asgard, that system will, via thermal forces, move to the Stargate. The zeros or o-points we will call the Goa'uld. If the system moves from the universe of an Asgard, at the Stargate, to the universe of a Goa'uld, that system will, via thermal forces, move to the Goa'uld. The Goa'uld is a long distance from the Stargate, and it will take a large amount of energy to reach it. The the Asgard will outnumber the Goa'uld by one. The cosmos can be closed by coupling the cosmological quantum field to a cosmological thermal bath with temperature T_c . A diagram of the cosmological quantum system is shown in Fig. 6. It should be noted that gravity is quantized at the cosmological scale by the quantization of the cosmological field.

The HST is closely related to methods in Machine and Deep Learning (MLDL). The HST starts with a convolution with a bank of complete, orthogonal, localized, harmonic functions (mother wavelets). This is followed by a nonlinear function that is roughly $\ln$. This is repeated for each order. Finally, a pooling operator is applied to sum

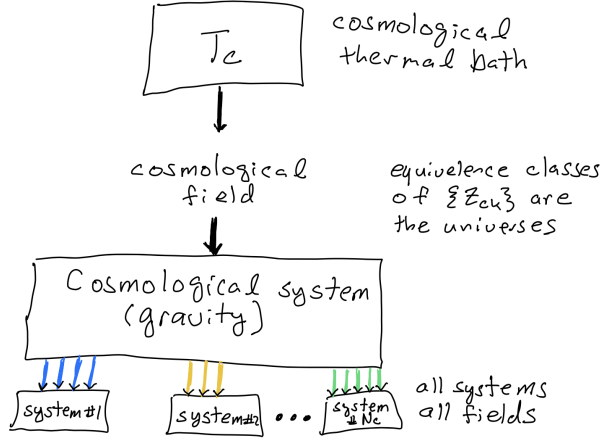


FIG. 6. Cosmological quantum system couples all fields from all systems.

over the partition of unity given by the complementary function (the father wavelet, to maintain harmonicity). This is then followed by a Principal Component Analysis (PCA, carefully constructed to respect the cyclic nature of the phase) to determine the low dimensional linear subspace of the Hilbert space to which the dynamics is constrained. A Singular Value Decomposition (SVD) of the initial fields and coupling constants to the HST of the output fields can be done at this stage to determine the solutions to the RGEs. Finally, a Multi Layer Perceptron (MLP) with ReLU activation (a piece-wise linear function approximator that can handle the singular nature of the mappings) with an encoder to action-angle coordinates (solutions of the Hamilton-Jacobi equations) then a decoder architecture can be applied to approximate the analytic function $H(\beta)$. In contrast to MLDL practice there are no parameters to estimate, compared to the 100 million or more in convolutional networks, which are searching over a Banach space. That is followed by the very fast shallow linear PCA and SVD analysis. The final deep step of the MLP has a pre-determined neural structure and is very fast to optimize since it is approximating an analytic (maximally flat) function. Because the forward HST was structured with a complete orthogonal basis, a fast inverse HST (iHST) has been constructed. Given the theory behind the HST, MLDL has been reduced to topological discovery.

This introductory discussion is concluded by summarizing this transformational methodology and putting it into historical context. The methodology is summarized in Fig. 7. The HST followed by the PCA transforms the motion from the cotangent bundle T^*M^n with symplectic metric governed by Hamilton's equation, to geodesic motion on C^n governed by the Cauchy-Riemann equations. The Hamiltonian functional $H[f, \pi]$ maps to the analytic function $H(\beta)$ that specifies the geodesics and adjoints, equivalently the topology of the dynamical manifold. The S-matrix or complex curvature is then given

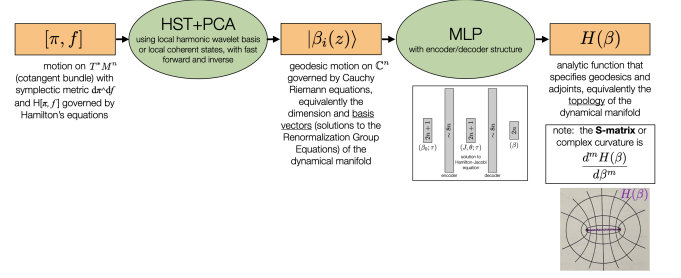


FIG. 7. The two step process for topological discovery of dynamical systems.

by the derivatives $d^m H(\beta)/d\beta^m$. The topological obstructions associated with the equivalence classes of the topology are the ground states of the dynamical system and the action of those ground states are the rest masses of the dynamics. The MLP approximates the key analytic function $H(\beta)$. The two steps in this diagram are a process of topological discovery. The first step of the HST+PCA discovers the dimension and basis vectors. The second step of the MLP approximates the analytic function $H(\beta)$ on this basis.

The concept of the S-matrix was first introduced by Werner Heisenberg in 1943 [8]. He lost interest in it, most likely, because he lacked concepts of analyticity. It was picked up again by Chew and Low [9] in 1955 and was most completely developed by Lev Landau [10] and Richard Cutkosky [11] by 1960. A good survey of this work was written by Chew [12]. This work was based on the concepts of analyticity and unitarity. It further used the expansion of an exponential generator employed by Richard Feynman in the path integral approach to quantum field theory. Furthermore, it evaluated the resulting integrals using a Fourier basis. It had no physical basis for the analyticity. Because of this, the theory was incomplete and did not form a well defined theoretical structure. It was also very difficult to calculate, and was not evaluated past second order. This line of research was abandoned in favor of the path integral formulation.

The work presented in this paper addresses these deficiencies. First, analyticity is the key to this work and has a deep physical basis coming from the isomorphism between Hamilton's equations and the Cauchy-Riemann equations. There is no need to enforce unitarity, since analyticity leads to unitarity. The exponential generator is replaced by a logarithmic generator so that the dynamics are constrained to a linear subspace. A wavelet basis with local support is used in place of the Fourier basis so that the integrals do not suffer from infinities, more simply said, they are compatible with the evaluation of integrals on manifolds. The analytic function $H(\beta)$ specifies the topology of the dynamical manifold.

This work is also closely related to the Wigner-Weyl transformation [13–15]. They wanted to transform from the cotangent bundle to $\mathbb{R}^n$, then solve an ODE. The problems were they: (1) used a global Fourier basis, (2)

only calculated to second order, and (3) did not include the logarithmic transform. This resulted in complicated corrections to the commutator, divergences in the evaluation, and an incomplete transformation.

The HST follows the basic philosophy of Eugene Wigner and Hermann Weyl of transforming from the cotangent bundle to $\mathbb{C}^n$, but uses an orthogonal local partition of unity based on coherent wavelet states to evaluate the integrals on the dynamic manifold, and embeds the complex logarithm in the transformation so that the dynamical manifold is aligned with $\mathbb{C}^n$, where n is the number of fields. The beauty of the HST is that the ODE is Laplace's equation (the Cauchy-Riemann equations) and the dynamical motion is simply geodesic motion, like for general relativity, given the topology of the dynamical manifold. Dynamics and Quantum Field Theory has been reduced to a matter of geometry — the geometry of physics [16].

This paper will first develop the theory of the HST in Sec. II, and the logical next theory of the coupling of quantum systems in Sec. III, building on the foundation of the first theory. A simple example, guiding center motion of an electron and an ion, with a closed form solution via canonical variable transformation is shown in Sec. IV. This example gives important physical insight into what the HST is doing. That will be followed with a discussion of how to build other group symmetries into the HST in Sec. VI. The physical interpretation of the first and second order moments will be explored in Sec. VII. Further insight will be given in Sec. VIII where the HST is applied to a 1D ϕ^4 theory. The practical implementation (a fast GPU based computation) of the HST is given in Sec. IX. We emphasize that this is a significant difference from the Lagrangian approach where there is no prescribed renormalization procedure that is guaranteed to work. This implementation is applied to the stochastic 1D Burger's equation in Sec. X, and to a planer 2D MHD pulsed-power implosion in Sec. XI. The conclusions and discussions will be given in Sec. XII.

II. THEORY OF THE HEISENBERG SCATTERING TRANSFORMATION (HST)

The transformation we endeavour to construct takes the dynamics to a manifold that is a linear subspace of a Hilbert space with basis vectors that are the solution to the Renormalization Group Equations (RGEs). In this subspace, the dynamics is geodesic motion with the topology given by the Hamiltonian function $H(z)$, an analytic function on $\mathbb{C}^n$. Another way of looking at this is that the motion is harmonic. We start by constructing a logarithmic generator of the function $\bar{H} \equiv iH$. First, Taylor expand the function $\bar{H}$ about z_0 giving

$$\bar{H}(z) = \sum_{n=0}^{\infty} \frac{1}{n!} S_m(z_0) (z - z_0)^n, \quad (3)$$

where

$$S_m(z) = \frac{d^m \bar{H}(z)}{dz^m} = \frac{dH_m}{dz}. \quad (4)$$

which serves as a definition of $H_m(z)$. Now we need to find an expression for H_m given H_{m-1} . Start by assuming that H_m is a functional of the field and the field momentum, that is

$$H_m(z(x)) = F[\pi(x), f(x)] \quad (5)$$

and define the wavelet transformation

$$\psi_p \star f(x) = \int \psi_p(x') f(x - x') dx' \quad (6)$$

where $\psi_p(x)$ are a normalized, orthogonal, localized and harmonic (that is coherent states) such that

$$\psi_p(x) \equiv p^2 \psi(px), \quad (7)$$

and

$$\phi_{px}(x') \equiv p^2 \phi(p(x' - x)), \quad (8)$$

where $\psi(x)$ and $\phi(x)$ are the Mother and Father wavelets that satisfy the Littlewood-Pauley condition. Consider

$$z_p(x) \equiv \psi_p \star H_{m-1}(z(x)). \quad (9)$$

It can be proven, using the fact that $\pi(x)$ and $f(x)$ satisfy the field equations (Hamilton's equations, that is a diffeomorphism on a cotangent bundle T^*M^n), $\psi_p(x)$ are coherent states, and the functional chain rule; that $z_p(x)$ is a trajectory on the complex plane that satisfies the Cauchy-Riemann conditions. That is to say, $z_p(x)$ is an analytic trajectory. We now want to calculate the analytic function that gives this trajectory. See Fig. 8 for a graphical representation of this derivation. Take the covector dz_p along this trajectory, change to radial coordinates about $z_0 = -e^{i \arg(z_p)}$ such that $z = z_p - z_0 = Je^{i\theta}$, and rotate by $\pi/2$ (that is multiply by i), to get these expressions for the covector

$$\begin{aligned} dH_m(z) &= i \frac{dz_p}{|z|} = i \left(\frac{dz_{\parallel}}{|z|} + i \frac{dz_{\perp}}{|z|} \right) \\ &= i \left(\frac{dJ}{J} + i\theta \right) = i d(\ln(Je^{i\theta})) \\ &= i d(\ln z) = d(i \ln z). \end{aligned} \quad (10)$$

Defining

$$R_0(z) \equiv z + e^{i \arg(z)}, \quad (11)$$

one gets

$$\begin{aligned} H_m &= i \ln z = i \ln(R_0(z_p)) \\ &= i \ln R_0 \psi_{p_m} \star H_{m-1}. \end{aligned} \quad (12)$$

We use $f(x)$ as shorthand for any $F[\pi(x), f(x)]$, start with

$$H_0 = i \ln R_0 f(x), \quad (13)$$

remove the explicit x dependence with a convolution with a final convolution with ϕ_{px} , choose normally ordered paths so that

$$p_{k+1} < p_k \text{ and } x_{k+1} > x_k, \quad (14)$$

define

$$p = \sum_{k=1}^m p_k \text{ and } x = \sum_{k=1}^m x_k, \quad (15)$$

and we get the following expression for the iterative logarithmic generating functional (the HST)

$$H_m[f(x)](z) = \phi_{px} \star \left(\prod_{k=0}^m i \ln R_0 \psi_{p_k} \star \right) i \ln R_0 f(x), \quad (16)$$

where $z \equiv p + ix$. Since this Logarithmic Generating functional (LG) which by construction generates $iH(z) = \bar{H}(z)$, not e^{iH} as is normally done in the Lagrangian approach with the exponential generator Z , it follows that

$$\text{LG}''(z) = 0 \quad (17)$$

which means that the transformation has flattened the space so that the infinitesimal generator, iH , is now the finite group generator. We have transformed to the basis $|\beta(p, x)\rangle$, the Hilbert space of all possible solutions to the RGEs, corresponding to $S_m = dH_m/dz$ the S-matrix. The motion will be confined to an n -dimensional complex linear hyperplane and will be geodesic motion on that submanifold, determined by the analytic function $H(\beta)$, with curvature $dH/d\beta = 1/\beta = S$. Streamlines of the motion (the H group action) are given by $\text{Re}(H(\beta)) = \text{constant}$, and streamlines of dH (the $\text{Ad}(H)$ group action) are given by $\text{Im}(H(\beta)) = \text{constant}$.

Now let's review what the HST is doing. The $\psi_p \star$ is generating $z_p(x)$ a parametric trajectory of the analytic function iH . The $\ln R_0$ conformal (canonical) transformation is flattening the space onto the cylinder by transforming into polar coordinates about R_0 . The i conformal transformation is rotating $\pi/2$ from the tangent direction to the gradient (covector) direction. Another perspective views it as a canonical transformation that is exchanging the field momentum with the field. The choice of R_0 ensures that the repeated application of the mapping converges to a circle of radius π about the origin on the cylinder. This is because R_0 is a fixed point of $\ln$, that is $\ln(R_0(z)) \rightarrow z$ as $|z| \rightarrow 0$. The convergence to the origin is exponential for large deviations, then logarithmic for small deviations.

It should be noted that the transformation is no longer stationary since the Father Wavelet only averages over as

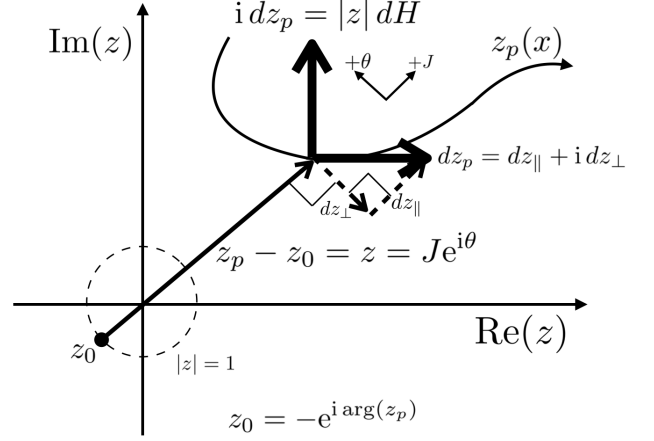


FIG. 8. Trajectory in $\mathbb{C}$ given by $z_p(x)$ projected onto polar coordinates about z_0 and rotated to the normal (dH covector) direction. See Eq. (12) for details.

large of a patch as it has to do. Nothing prevents this partition of unity from being summed over a larger domain in x , if the process is stationary over that domain. This transformation is complex from the beginning to the end. The connection to previous work can be seen by examining the limiting behavior of the $\ln(R_0(z))$ mapping

$$\begin{aligned} \ln(R_0(z)) &\xrightarrow{|z| \rightarrow 0} z \\ &\xrightarrow{|z| \rightarrow \infty} \ln |z| \quad (\text{our previous work}) \\ &\xrightarrow{|z| \rightarrow \infty} \ln |z| e^{i \arg(z) / \ln |z|} \\ &\sim \sum_{k=0}^n |z| e^{i k \arg(z)} \quad (\text{that is, WPH}) \\ &\xrightarrow{n \rightarrow 0} |z| \quad (\text{that is, MST}). \end{aligned} \quad (18)$$

The characteristic of the fixed point at the origin is the statement of the first limit. The second limit is effectively what has been used in our previous work [17]. In the third limit (which keeps the small $\arg$ imaginary term), the $\ln$ chirps the pulse, generating the harmonics that are explicitly generated in the fourth line by Wavelet Phase Harmonics (WPH) [18–21]. WPH is the attempt by Stéphane Mallat and his collaborators to incorporate phase in the transformation. The original Mallat Scattering Transformation (MST) [1] is just using the first term in that series.

Let us consider the simplest case of a set of poles and zeros, $\{\beta_{0k}\}$. The motion can now be identified by β_{0k} and $\Delta H_k = E - H_{0k}$ and a cyclic variable θ . Now consider that the state of the system is observed at τ_0 , then again at $\tau_0 + \Delta\tau$. The system will have a characteristic frequency near β_{0k} (that is $|\beta - \beta_{0k}| \ll 1$) of $\omega_0 = dH_2/d\beta$. If $\Delta\tau \lesssim 1/\omega_0$, then $\Delta\theta$ can be estimated. This is the classical limit. Things are very different if $\Delta\tau \gg 1/\omega_0$. Although the motion is constrained to the

submanifold and more specifically to a leaf of that submanifold, small errors in estimating β_0 will give large errors in estimating both the leaf and $\Delta\theta$. The nonlinear curvature will lead to very large differences in $\Delta\theta$, so that the system will remain close to the same leaf, but $\Delta\theta$ will not be able to be determined. Furthermore, it can not be determined how many times it went around. This means that $\Delta\omega_i = 1/2\pi n_i$. The ω here comes from the solution to the Hamilton-Jacobi equations so that $H = H(J)$, $dJ/dt = -\partial H/\partial\psi = 0$ (which implies that both J and $H(J)$ are constants), and $d\psi/dt = \partial H/\partial J = \omega(J) = \text{constant}$ so that $\psi = \psi_0 + \omega(J)t$. The outcome is that H is quantized such that $H = H_0 + \Delta H_n$ where $\Delta H_n = \Delta H(n_i)$.

Pause to consider what this means. The system is quantized in terms of the frequency of its ground state, not in the frequency of the ground state of the hydrogen atom. This modifies the uncertainty principle to be set to a characteristic action of the system $H_0/\omega(H_0)$, not by $\hbar$. This can be simply viewed by looking at the expression of the Cauchy-Riemann conditions in terms of the $\bar{p}$ and $\bar{q}$ coordinates as $\{\bar{p}, \bar{q}\} = i$, where the i has appeared to take the vector to the covector on $\mathbb{C}$. There is no $\hbar$ in this expression. The uncertainty in the determination of $\bar{p}$ and $\bar{q}$ is in terms of the characteristic action of the system. This comes about because determination of the covector $\bar{p}$ is a differential measurement of the coordinate $\Delta\bar{q}$ over $\Delta\tau$. In order to increase the accuracy of the measurement of the covector, $\Delta\tau$ must be increased, but there is a limit to how large that $\Delta\tau$ can be and still get an accurate measurement of the coordinate. So there is a lower bound to the determination of the product of the uncertainties of canonically conjugate variables set by the characteristic action of the system, not by $\hbar$. The error in the measurement of the initial coordinate does not matter because the uncertainty in the prediction of its position is growing exponentially as a function of $\omega\Delta\tau$.

A final consideration is the size of the HST as a function of the scale of the stationary of the process. Take an N^{N_D} image that is stationary on a patch of size $(N/M)^{N_D}$. The minimum number of patches will be for a stationary process with $M = 1$. The completely non-stationary process will have the maximum number of patches with $M = N$. A general expression for the number of coefficients can be written as

$$N_c = M^{N_D} - 1 + \frac{(\log_2 N)^{N_D}}{N_D!} \quad \text{in } N_D \text{ dimensions.} \quad (19)$$

The completely non-stationary process will have the same number of coefficients as the original image N^{N_D} . The stationary process will have logarithmically fewer $(\log_2 N)^{N_D}$.

Go to circle of radius π . Discuss that the convolution decomposes on the basis of the scale of averaging and generates an analytic trajectory on the complex plane. The $\ln$ changes to polar coordinates after translating the origin to a fixed point of the mapping, all canonical transformations. This flattens the space, and exposes the

curvature of the manifold as the expansion coefficients $dH/d\beta$. Note that H is defining the Poincaré one form $\pi \wedge df + iH \wedge dx$, the exterior derivative is taken to get the S-matrix and curvature two form, that is the Poincaré two form.

III. QUANTIZING SYSTEMS AND COUPLING SYSTEMS BY A SECOND QUANTIZATION

We start with a closer examination of the Weyl-Heisenberg group. Conventionally, this is a two parameter group written as

$$\mathbb{H} = H_{H[\pi, f]}^1 \otimes \mathbb{R}_\tau^1 \quad (20)$$

with the symplectic metric $d\pi \wedge df$. The group parameter of the one dimensional trivial group $\mathbb{R}_\tau^1$ is $\tau = \sqrt{t^2 - x^2 - y^2 - z^2}$ (which is t for a classical dynamical system, and $\sqrt{x^2 + y^2 + z^2}$ for a system in stationary statistical equilibrium like an Ising model). The non-trivial topology of the group is currently hidden in $H_{H[\pi, f]}^1$. The topology of τ is not flat and can only be locally mapped onto $\mathbb{R}$. In fact, the topology of τ will be shown to be complementary to the topology of H , formally adjoint. To expose the topology, first extend phase space with the symplectic metric $d\pi \wedge df + dH \wedge d\tau$ and express the Weyl-Heisenberg group as

$$\mathbb{H} = H_{H[\pi, f]}^1 \otimes dH_\tau^1. \quad (21)$$

Next, transform $H[\pi, f]$, a functional on the cotangent bundle T^*M^n where n is the number of fields, to $H(\beta)$, an analytic function on $\mathbb{C}^n$, using the HST, a logarithmic generating functional given in Eq. 16. The Weyl-Heisenberg group can now be expressed as

$$\mathbb{H} = H_{\text{Re}(H(\beta))}^1 \otimes dH_{\text{Im}(H(\beta))=\tau}^1 = H \otimes \text{Ad}(H). \quad (22)$$

Because $H(\beta)$ is analytic, the Cauchy-Riemann equations (Kramers-Kronig Relations or Hamilton's Equations) can be solved for $\text{Im}(H(\beta))$ given $\text{Re}(H(\beta))$. The non-trivial topology of the Weyl-Heisenberg group is now exposed and specified by $H(\beta)$ with curvature $d^m H/d\beta^m$ (the S-matrix). This topology can be alternatively specified by the equivalence or homology classes given by the ground states, β_{0k} , (that is the topological obstructions or the singularities or the topological indexes) of $H(\beta)$. The β_{0k} are essentially a boundary condition, and $H(\beta)$ is the solution of Laplace's Equation given that boundary condition. The perspective on dynamics given by this formalism is exactly that of general relativity, where all motion is geodesic motion with the force coming from a curvature of space (that is $d^m H/d\beta^m$) induced by the topology of the $H(\beta)$ function.

The relationship to the Werner Heisenberg's 1943 S-matrix ideas comes from the fact that the S-matrix or curvature is $d^m H/d\beta^m$, so we are calculating the S-matrix by calculating $H(\beta)$. Just take the derivative.

This is why Eq. 16 is called the Heisenberg Scattering Transformation. Second, the HST is also closely related to the ideas of Eugene Wigner and Hermann Weyl, when they constructed the Wigner-Weyl transformation. They wanted to transform from the cotangent bundle to $\mathbb{R}^n$, then solve an ODE. The HST follows this basic philosophy by transforming from the cotangent bundle to $\mathbb{C}^n$, then solving Laplace's Equation.

This is where things deviate from traditional quantum theory, but make a lot more sense. Simply, primary quantization is quantization of the H group action yielding the particles of the QFT (or dynamical system), and secondary quantization is quantization of the $\text{Ad}(H)$ group action yielding the gauge bosons of the QFT (or dynamical system) that couples the QFT to other systems (that is statistical mechanics or kinetic theory, through which dissipation and statistical equilibrium with a heat bath can be introduced). The process of quantization simply comes from enforcing a periodic boundary condition on the probability in action-angle coordinates (that is around a cycle), once the motion becomes chaotic and must be treated statistically.

The conventional introduction of $\hbar$ is not fundamental to the quantization but is a limit on the space like observation of τ in relativistic systems. The difference between quantum (statistical) observation and classical (deterministic) observation is whether the time between observations is long enough to allow the system to become ergodic (in nonlinear dynamics terminology) or decorrelate (in quantum mechanics terminology). When a system is moving at relativistic velocity, the system can never be observed classically in a space-like interval, that is a rest frame. This is the traditional $\hbar$ uncertainty limit. In non-relativistic systems it is possible to observe the system classically in a rest frame. If the system is observed quantum mechanically, that is statistically, there is a natural uncertainty limit associated with the action of the ground state, not the action of relativistic motion. Note that even for relativistic systems, a classical observation can be made in the frame of the particle, but not in a stationary space-like frame. This is the origin of the quantum paradoxes like the EPR paradox. These paradoxes should be logically disturbing since A and not A is false, that is the theory is not valid. These paradoxes are resolved with this quantum paradigm when it is recognized that there are distinct classical and quantum limits.

Finally, when one considers a system in equilibrium so that the probability is

$$p(\beta) \sim e^{iH(\beta)} = e^{i\text{Re}(H(\beta)) - \text{Im}(H(\beta))}, \quad (23)$$

the relationship between the primary quantization of the system (that is QFT or H) with the factor i , and the secondary quantization of the system to an external system (that is statistical mechanics or dH) with the factor of -1 is apparent.

This is the coupling theory. Start by very carefully and slowly describing the new paradigm on quantization using the Δt thought experiment. Very carefully say that

systems are quantized at a characteristic scale $\hbar$ action, that is characteristic action of the poles and zeros of the system, not at a universal scale $\hbar$. Emphasize the connection to nonlinear dynamics, statistical physics, kinetics, and continuum fluid dynamics. Then carefully describe the second quantization for subordinate and superordinate coupling. Then talk about how this is the basic Lego block for systems of systems. Then introduce gravity as the ultimate system of systems with a force of harmony and simplicity. Then give examples of different systems and the scale at which they are quantized.

IV. SIMPLE EXAMPLE GIVING PHYSICAL INSIGHT TO HST

This is the GCA example. Show the level diagram.

V. INCLUSION OF DISSIPATION IN THE DYNAMICS

The inclusion of a dissipative term in the dynamics is related to using an extended Hamiltonian, but is nuanced, and can be understood much better using this theory. The inclusion of a dissipative term in the system of PDEs adds another "coupling constant" to the dynamics which can be varied and the PCA associated with that variation determined. Coupling constant is put in quotes because this direction in the Hilbert space is in the covector plane dH , not the covector plane dC . It is foliating the $\mathbb{C}^n$ space of the second quantization dH for superordinate coupling, not foliating the $\mathbb{C}^{n_c}$ space of second quantization dC for subordinate coupling. It is not associated with a subordinate coupling being driven by this system but is the result of a superordinate system driving this system. In this way the dynamics is extended into one component of the dH plane. The nuance is that this extended direction will not generate another covector component for superordinate coupling, it is the use of an existing covector component.

VI. BUILDING IN ADDITIONAL GROUP SYMMETRIES

There is nothing special with respect to the Hamiltonian portion of the Weyl-Heisenberg group H^1 . The procedure can be repeated with any other group symmetry of the dynamics $G_k^{m_k}$. Form the cotangent using the group action and its adjoint with a symplectic metric. Use this to foliate the space. This can continue up to the point that

$$n = 1 + \sum_k^{N_g} n_k, \quad (24)$$

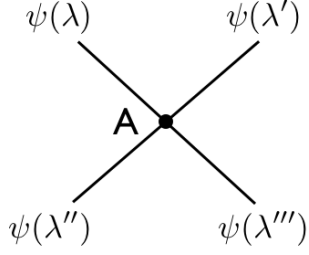


FIG. 9. Four point scattering vertex of the ϕ^4 theory, where $A(\lambda, \lambda', \lambda'', \lambda''')$ is the four point vertex scattering strength.

when the motion becomes integrable, that is fully foliated. The dynamics is now governed by the product group

$$H^1 \otimes G_1^{n_1} \otimes G_2^{n_2} \otimes \cdots \otimes G_k^{n_k} \otimes \cdots \otimes G_{N_g}^{n_{N_g}}. \quad (25)$$

The path variable is now extended to include a path for each of the additional groups so that

$$p = (p_H, p_{G_1}, p_{G_2}, \dots, p_{G_{N_g}}). \quad (26)$$

This is simply an expression of Noether's Theorem, that there are additional constants of the motion for each additional symmetry.

VII. PHYSICAL INTERPRETATION OF THE FIRST AND SECOND ORDER HST FROM THE LAGRANGIAN PERSPECTIVE IN THE QUANTUM LIMIT

Need to understand these terms as they relate to the $1/\omega_0$ and the $S_0 = H_0/\omega_0$

VIII. EXAMPLE OF 1D ϕ^4 FIELD THEORY

Rework ϕ^4 theory using new paradigm and show the $m \geq 5$ terms are zero and diagram

IX. IMPLEMENTATION OF THE HST

This is the implementation. Practically how it is done, including MLDL PCA and MLP.

X. APPLICATION TO THE 1D BURGER'S EQUATION

This is it applied to stochastic burgers.

XI. APPLICATION TO 2D MHD

This is it applied to MHD.

XII. CONCLUSIONS

Review what has been done. More cosmology to come. Discuss how this relates Quantum Field Theory to statistical mechanics to nonlinear dynamics to AI.

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